

3. Welsh Water is responsible for supplying and maintaining safe water supplies in Wales. It monitors the composition of our drinking water.

The table below shows the composition of stream water supplying a reservoir in Wales. The samples were taken from three different streams.

Component	Concentration (units)		
	stream 1	stream 2	stream 3
sodium ions (Na^+)	5.0	5.0	5.0
potassium ions (K^+)	6.3	6.2	6.2
magnesium ions (Mg^{2+})	15.3	50.4	60.5
calcium ions (Ca^{2+})	50.4	50.1	101.1
chloride ions (Cl^-)	30.1	30.0	30.2
hydrogen carbonate ions (HCO_3^-)	10.1	50.1	40.0
dissolved sulfur dioxide	9.9	12.3	15.0
dissolved carbon dioxide	0.6	0.9	1.6
pH	5.7	5.5	5.4

Use the information in the table and your knowledge to answer the following questions.

- (a) Explain how human activity has affected the acidity of the streams. [3]

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- (b) A technician thinks the water in stream 1 is the softest and in stream 3 is the hardest.

Explain whether you agree. [3]

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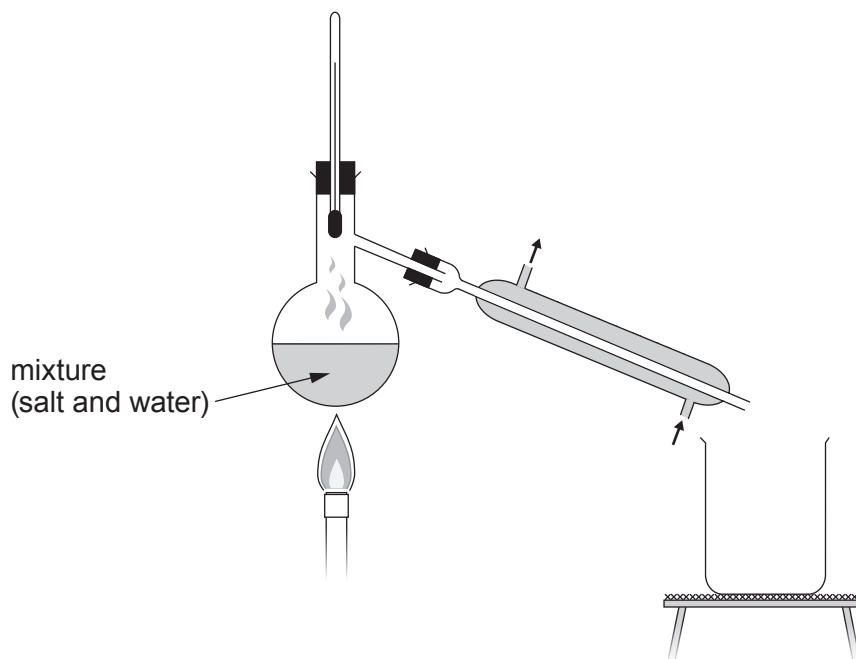
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- (c) Seawater can be desalinated using the apparatus below.



- (i) Describe how pure water is extracted using the apparatus above.

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- (ii) Desalination can be used to increase the water supply for the public in parts of the world where water is scarce.

Describe the factors a government must consider when investing in industrial-sized desalination plants.

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